

CNC-X100 Technical Specifications & Engineering Data

Product: CNC Machine | Model: CNC-X100 | Document Type: TECHNICAL_SPECIFICATION

1. Machine Dimensions & Physical Parameters

- Machine Category: 3-Axis Vertical Machining Center.
- Machine Footprint: 2,800 mm Width x 2,250 mm Depth x 2,650 mm Height.
- Total Machine Weight: 4,200 kg (9,260 lbs).
- Table Dimensions: 900 mm x 450 mm.
- Maximum Workpiece / Table Load Capacity: 600 kg (1,320 lbs).
- T-Slot Configuration: 5 slots, 18 mm width, 100 mm pitch.

2. Electrical & Power Requirements

- Input Voltage: 3-Phase AC 400V $\pm$ 10%, 50/60 Hz.
- Total Connected Load: 35 kVA.
- Full Load Current: 48 Amps.
- Recommended Circuit Breaker: 63 Amp 3-pole breaker with RCCB.
- Control Voltage: 24V DC regulated power supply.
- Grounding Requirement: Dedicated earth ground rod with ground resistance <5 Ohms.

3. Operating Temperature & Environmental Limits

- Operating Temperature Range: +15°C to +35°C (+59°F to +95°F).
- Storage Temperature Range: -10°C to +50°C.
- Relative Humidity: 30% to 80% non-condensing.
- Maximum Altitude: 1,000 meters above sea level without power derating.

4. Spindle Unit & Tooling Specifications

- Spindle Taper: BT40 (ISO 7388-1).
- Spindle Speed Range: 100 RPM to 12,000 RPM (Direct Drive).
- Spindle Motor Power: 11 kW continuous / 15 kW 30-min peak duty.
- Maximum Spindle Torque: 95 Nm @ 1,500 RPM.
- Automatic Tool Changer (ATC): 24-tool carousel magazine.
- Tool-to-Tool Change Time: 1.8 seconds.
- Tool Retention Stud Torque: 45 Nm.

5. Motion & Axis Parameters

- X-Axis Travel: 800 mm | Y-Axis Travel: 450 mm | Z-Axis Travel: 500 mm
- Rapid Traverse Rates: X: 30 m/min, Y: 30 m/min, Z: 24 m/min.
- Maximum Cutting Feed Rate: 10,000 mm/min.
- Positioning Accuracy (ISO 230-2): -0.005 mm.
- Repeatability (ISO 230-2): -0.003 mm.

6. Pneumatic, Coolant & Lubrication Specifications

- Pneumatic Pressure Range: 6.0 bar to 8.0 bar (nominal 6.5 bar). Air flow: 250 L/min.
- Coolant Tank Capacity: 200 Liters.
- Recommended Coolant: 6% water-soluble synthetic coolant emulsion.
- Normal Coolant Fill Level: 80% fill mark (Minimum acceptable level: 50%).
- Coolant Pump Pressure & Flow: 3.5 bar @ 35 L/min.
- E105 Alarm Threshold: Flow rate sensor FM-1 detects below 8.0 L/min or thermal relay Q4 trips.
- Central Lubrication Oil: ISO VG 68 slide-way oil (2.0-Liter tank).

7. Communication & Telemetry Protocols

- Ethernet Configuration: Dual RJ45 1000BASE-T ports. Static IP: 192.168.1.150
- Supported Protocols: Modbus TCP/IP (Port 502), OPC UA Server (Port 4840), MTConnect Agent (Port 5000).